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Research in Sports Medicine
An International Journal

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/gspm20

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To cite this article: Reid A. Syrydiuk, Adrian J. Boltz, Landon B. Lempke, Kathryn van Pelt, Vinodh Balendran, Katie Scott, Michael Vesia, Bruno Giordani, Michael A. McCrea, Thomas W. McAllister, Steven P. Broglio & CARE Consortium Investigators (22 Jan 2025): Association between magnetic resonance imaging use and recovery time following concussion: a CARE consortium study, Research in Sports Medicine, DOI: 10.1080/15438627.2025.2453684

To link to this article: https://doi.org/10.1080/15438627.2025.2453684

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Published online: 22 Jan 2025.

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RESEARCH IN SPORTS MEDICINE
https://doi.org/10.1080/15438627.2025.2453684

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Association between magnetic resonance imaging use and recovery time following concussion: a CARE consortium study

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ABSTRACT

Researchers may implement magnetic resonance imaging (MRI) to evaluate neurophysiological metrics (e.g. connectivity) in athletes with sports-related concussion (SRC). MRI usage has been purported to exacerbate symptoms that mimic SRC symptomatology, possibly influencing recovery. The present study examined MRI use on SRC recovery outcomes in collegiate athletes. Ninety student-athletes (45 with MRI acutely post-SRC, 45 non-MRI) from four collegiate sites were analysed. Multivariable Cox proportional hazard regressions were used to measure the SRC recovery risk [return-to-play (RTP) initiation (RTP-I) and unrestricted RTP] with hazard ratios (HR) and 95% confidence intervals (95% CI). No hazard differences for median days to RTP-I [MRI: 10, interquartile range (IQR) = 7–15; non-MRI: 7 (IQR = 4–9)] were observed (HR = 0.75; 95% CI: 0.28–1.21). No hazard differences for median days to RTP [(MRI: 17 (IQR = 11–60); non-MRI: 12 (IQR = 8–24)] were observed (HR = 0.77; 95% CI: 0.29–1.26). MRI use post-SRC does not affect recovery; researchers/clinicians can employ such a procedure without affecting athlete recovery.

ARTICLE HISTORY
Received 3 May 2023
Accepted 8 January 2025

KEYWORDS
Sport-related concussion; magnetic resonance imaging; return-to-play; sports medicine

Introduction

Sports-related concussion (SRC) is "a traumatic brain injury induced by biomechanical forces", associated with impairments in physical, neurological, and physiological domains (Borich et al., 2015; Kamins et al., 2017; Manley et al., 2017; McCrory et al., 2017). SRC assessment commonly includes a multi-faceted approach involving neuropsychological testing, self-reported symptoms, and a balance examination that supports the clinical examination (McCrory et al., 2017). In National

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Collegiate Athletic Association (NCAA) athletes, recovery from SRC is often characterized by a progressive resolution of symptoms and returning to activities such as school, work, or physical activity (Aderman et al., 2022; McCrory et al., 2017). Patients recovering from SRC are often asked which and to what extent they are experiencing certain symptoms through standardized symptom assessments (Lempke et al., 2020, 2022). Numerous studies have reported that the total acute symptom burden following SRC is one of the strongest predictors of increased recovery time in athletes (Iverson et al., 2017). SRC symptoms typically resolve within 1 week, with the median return to play (RTP) clearance achieved within 2 weeks (Broglio et al., 2022).

Neuroimaging via magnetic resonance imaging (MRI) is increasingly being implemented by researchers to explore structural and functional neural changes after SRC, with prior work showing both function and structure associated with reported symptoms following SRC and acute post-injury neurocognitive testing (Klein et al., 2019; McCrea et al., 2017; Yuh et al., 2013). Although a non-invasive technique, MRI can provoke increased patient anxiety during the procedure (Yakar & Pirinçci, 2020). Anxiety has been reported in over 20% of athletes following SRC and is associated with delayed SRC recovery, making it a concern for SRC management that may be further exacerbated by neuroimaging (D'Alonzo et al., 2022). The known risk of undergoing an MRI and psychological symptom provocation may be an important consideration when developing research protocols for SRC recovery (Dewey et al., 2007; Yakar & Pirinçci, 2020). To our knowledge, the effect of MRI use on SRC recovery has yet to be examined and warrants exploration to determine if MRI poses a deleterious effect.

While the association between SRC and MRI findings has been studied, the influence of the MRI procedure itself on SRC recovery has not. Thus, the aim of the present study was to examine the association between MRI use and days to return to play (RTP) protocol initiation (RTP-I) and unrestricted RTP among collegiate student-athletes experiencing SRC.

Methods

The present study included student-athletes enrolled in the NCAA Department of Defense (DoD) Concussion Assessment, Research, and Education (CARE) Consortium. Briefly, the CARE Consortium recruited both collegiate athletes and military service academy members from 30 sites across the United States, who were evaluated pre- and post-SRC to help inform clinical care, natural history, and other crucial domains regarding SRC as described elsewhere (Broglio et al., 2017). Student-athletes enrolled in CARE between Fall 2014 and Spring 2020 were used for the present study. A subset of six CARE sites comprised the Advanced Research Core (ARC), which implemented additional assessments, including neuroimaging (i.e., MRI) at four of six sites among athletes who participated in high-contact sports (Broglio et al., 2017). Thus, athletes from the four ARC institutions were examined in the present study. The United States Army Medical Research and Materiel Command Human Research Protection Office and the Medical College of Wisconsin Institutional Review Board reviewed all study procedures. All student-athletes provided written informed consent prior to participation.

Concussed student-athletes

Only the first SRC an athlete sustained during the study period was included in the analyses. Student-athletes were excluded if they were non-NCAA athletes (i.e., non-varsity military service academy cadets) and varsity athletes with a non-SRC (e.g., falling on the sidewalk), or missing data for any of the following characteristics: prior SRC(s), sex, injury context (e.g., competition or practice/training), or site. Athletes who consented to the procedure had research MRI assessments conducted within 72 h of sustaining SRC. Assessments lasted approximately 1 h, and findings were not divulged to student-athletes or medical providers unless they yielded clinical importance (i.e., surgical intervention or medical disqualification required). Greater details of MRI procedures, including types of images collected and data analysis, have been delineated elsewhere (Broglio et al., 2017). We acknowledge the important role MRI plays in identifying clinically abnormal findings that may seriously jeopardize health, such as intracranial haemorrhaging (Bonney et al., 2020). Klein et al. has examined prevalence of significant (i.e., clinically abnormal) imaging findings within the CARE Consortium, reporting less than 1% of SRCs having associated abnormal findings (Klein et al., 2019). Notably, the present study did not aim to characterize functional or structural neurological outcomes from SRC in this cohort, only to determine the effect of the MRI procedure itself on SRC recovery.

Control student-athlete propensity matching

Propensity score matching was used to individually match concussed student-athletes who completed an MRI as part of the ARC protocol to concussed student-athletes who did not receive an MRI. Propensity scores were calculated based on the probability of MRI status as a function of cumulative covariate presence (sex, prior SRC(s), injury situation, and site) (Kane et al., 2020). Within the R Project for Statistical Programming (v4.0.3, Murray Hills, NJ), the "matchit" package was used in conjunction with the nearest neighbour method (Greifer & Stuart, 2021; Ho et al., 2025).

Assessments and outcomes

Following the SRC diagnosis, post-injury assessments were conducted at the following time points: immediate post-injury (i.e., <6 h), 24–48 h post-injury, upon athletes' initiation of RTP protocol (RTP-I), and finally, when the athlete was cleared for unrestricted RTP (URTP). Control student-athletes completed assessments at the same time intervals as the concussed athletes. Demographic information was collected via a Case Report Form (CRF) unique to each athlete (Broglio et al., 2017). The Sport Concussion Assessment Tool (SCAT) symptom checklist, a 22-item, 0–6 Likert scale questionnaire where athletes rate specific symptoms, was used to assess SRC symptom burden via the total symptom severity (TSS) score (range: 0–132). One of the SCAT TSS symptoms, "feeling nervous/anxious", was independently included in the analyses. The Brief Symptom Inventory-18 (BSI-18) was administered, with the anxiety portion (aggregate score of six questions, 0–4 Likert scale) and total Global Severity Index [(GSI); 0–72 score summing depression, anxiety, and somatic scores)] included in the analyses (Meachen et al., 2008).

In the present study, days of RTP-I and URTP were the two primary outcomes. The medical staff provided clearance to begin the RTP protocol following SRC diagnosis and guide restrictions from physical (e.g., team practices or competition) and cognitive (e.g., coursework) activities per CARE protocol (Broglio et al., 2017). Once all stages of the RTP protocol were completed, URTP clearance was granted by the local medical staff (Broglio et al., 2017).

Statistical analyses

Student-athlete characteristics (by MRI status and overall) are described using counts and proportions in Table 1. Group differences on the SCAT Nervous/Anxious rating (0–6) by MRI group at both baseline and post-injury time points were evaluated using Fisher's exact test, and BSI-18 Anxiety Raw Score (0–24) score differences were evaluated using Wilcoxon rank sum test. Overall SCAT TSS and BSI-18 GSI total score (0–72) were compared by MRI status using Wilcoxon rank sum test. Scores collected <24 h after initial injury were used for SCAT TSS (and Feeling Nervous/Anxious) and BSI-18 (GSI and Anxiety raw) scores.

Survival analyses were used to examine differential SRC recovery trajectories. Censoring at 30 and 60 days for RTP-I and URTP (respectively) was decided as in a previous study, over 90% of NCAA student-athletes achieved RTP-I by 30 days, and over 80% were at URTP within 60 days (Broglio et al., 2022). Univariate models examining trajectory differences on recovery (days to RTP-I and days to URTP separately) were used via a Log-rank test by sex, site, injury situation, previous SRC, and MRI status. SCAT TSS (discrete variable) on recovery was evaluated using Cox proportional hazard model. MRI status and days to RTP-I and URTP were evaluated separately using multivariable Cox proportional hazard models covaried for SCAT TSS with hazard ratios (HR) and 95% confidence intervals (CI) reported. SCAT TSS was included as a covariate due to its robust association with SRC recovery (Iverson et al., 2017). Model-associated Schoenfeld residual plots were visually assessed for any irregular patterns (i.e., non-horizontal) and provided sufficient evidence that the proportional hazards assumption was upheld. No other covariates were considered because other confounders (sex, prior SRC(s), injury situation,

Table 1. Student-athlete characteristics.

Characteristic	No MRI, $n = 45$[1]	MRI Done, $n = 45$[1]	Overall, $n = 90$[1]
Sex			
Female	8 (18%)	8 (18%)	16 (18%)
Male	37 (82%)	37 (82%)	74 (82%)
Previous diagnosed concussion	14 (31%)	14 (31%)	28 (31%)
Injury Situation			
Practice/Training	31 (69%)	31 (69%)	62 (69%)
Competition	14 (31%)	14 (31%)	28 (31%)
Site			
Site 1	11 (24%)	11 (24%)	22 (24%)
Site 2	12 (27%)	12 (27%)	24 (27%)
Site 3	7 (16%)	7 (16%)	14 (16%)
Site 4	15 (33%)	15 (33%)	30 (33%)
With Return to Play Initiation Event (not censored)	38 (84%)	40 (89%)	78 (87%)
With Unrestricted Return to Play Event (not censored)	35 (78%)	35 (78%)	70 (78%)

[1]n (%)

and site) were already accounted for via propensity score matching. All models were calculated such that HR > 1.0 indicated a higher instantaneous risk for recovery relative to the referent and HR < 1.0 represented a lower instantaneous risk for recovery relative to the referent. HR with 95% confidence intervals (95% CI) not including 1.0 were deemed statistically significant. All statistical analyses were conducted using the R Project for Statistical Programming (v4.0.3, Murray Hills, NJ), with $\alpha = 0.05$ *a priori*.

Results

Following the application of the exclusion criteria, 2,072 concussed student-athletes remained (see Figure 1). A total of 45 student-athletes had an MRI, with 45 student-athletes without an MRI successfully matched by propensity scores. Of the 90 included student-athletes, 18% were female, 31% had prior SRC, 69% of the SRCs occurred during practice/training, and occurred relatively equally across the four ARC sites (Table 1). At the baseline, no significant differences were observed for SCAT Nervous/Anxious rating ($p = 0.64$) or BSI-18 Anxiety Raw scores ($p = 0.72$) between MRI groups (see Table 2). Furthermore, neither the SCAT TSS ($p = 0.10$) nor BSI-18 GSI ($p = 0.20$) was different at the baseline between MRI groups. Following SRC, student-athletes with an MRI had significantly higher anxiety scores than the non-MRI group (Table 2) on the SCAT Nervous/Anxious rating ($p = 0.04$). No MRI group differences were identified for BSI-18 Anxiety scores at post-injury ($p = 0.46$). Student-athletes in the MRI group had significantly higher TSS scores than non-MRI student-athletes ($p = 0.01$) post-injury, and BSI-18 GSI scores did not differ between groups ($p = 0.20$).

Time to RTP protocol initiation

Approximately 13% (12/90) of the student-athletes did not progress to RTP-I at 30 days and were censored. Median days to RTP-I for the non-MRI and MRI groups were seven (IQR: 4–9) and ten (IQR: 7–15), respectively. Univariate analyses demonstrated no significant associations with days to RTP-I for site, sex, injury situation, prior SRC, or MRI status (p's > 0.05). No significant relationship between MRI use and days to RTP-I was observed (HR = 0.75; 95% CI: 0.28–1.21; Table 3) after controlling for SCAT TSS in the Cox proportional hazards model (Figure 2).

Time to unrestricted RTP

Approximately 22% (20/90) of the student-athletes did not progress to URTP within 60 days and were censored. Median days to URTP for the non-MRI and MRI groups were 12 (IQR: 8–24) and 17 (IQR: 11–60), respectively. Univariate Log-rank tests demonstrated sex, prior SRC, site, MRI status, and injury situation were not associated with days to URTP ($p > 0.05$). MRI status on days to URTP was not significant (HR = 0.77; 95% CI: 0.29–1.26; Table 3) after controlling for SCAT TSS in the Cox proportional hazards model (Figure 3).

Discussion

The present study aimed to compare SRC recovery timelines between student-athletes who received an MRI scan post-SRC and those who did not. After controlling for post-SRC

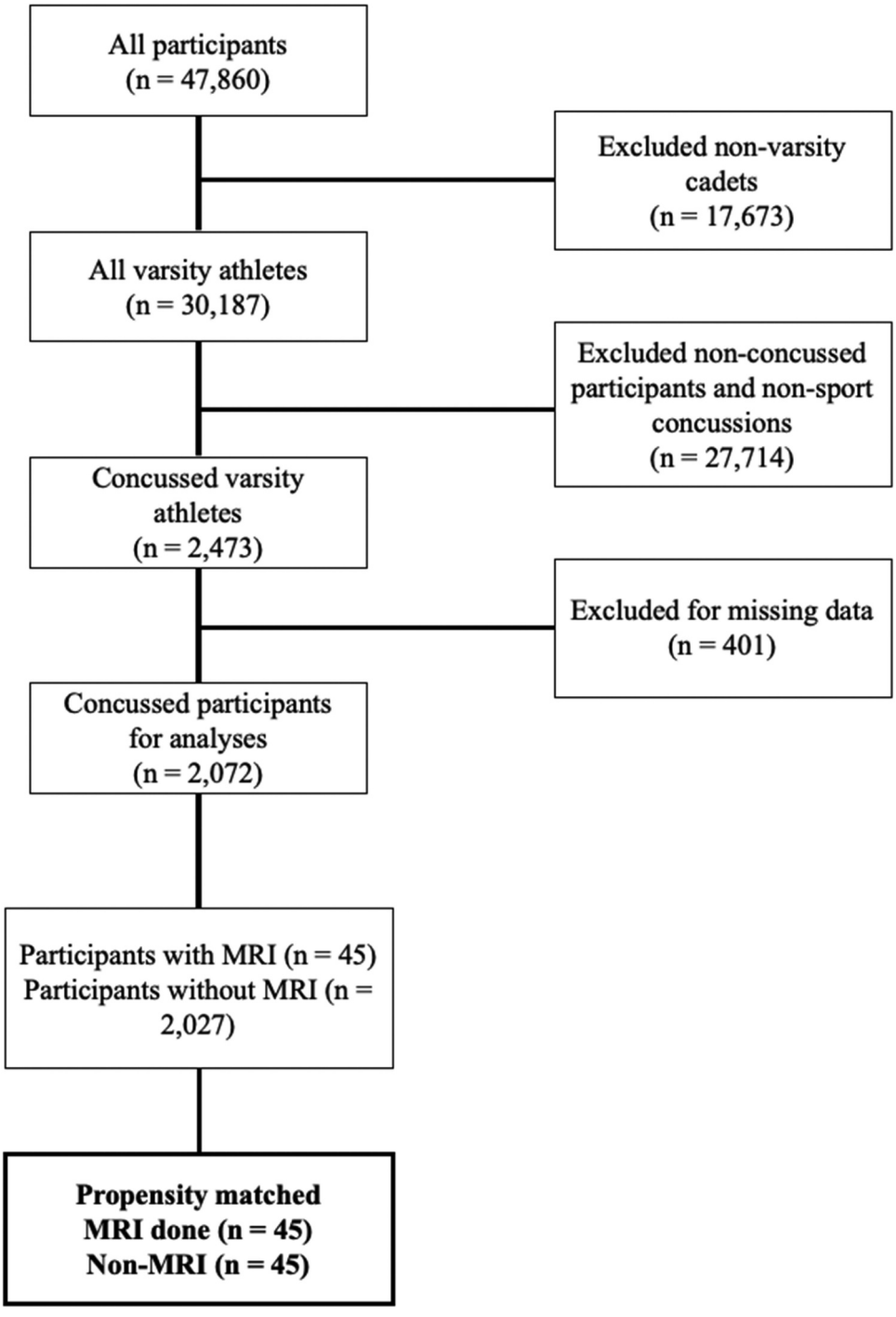

Figure 1. Student-athlete selection flowchart.

Table 2. Time to recovery and self-reported symptom burden (SCAT & BSI-18).

	Non-MRI	MRI Done	p-value†
SCAT Nervous/Anxious			
Baseline			0.64
Median (IQR)	0 (0–0)	0 (0–0)	
Mean (SD)	0.14 (0.41)	0.48 (1.17)	
Missing (n)	1	1	
Post-injury			0.04*
Median (IQR)	0 (0–1)	0 (0–1)	
Mean (SD)	0.62 (1.30)	0.82 (1.39)	
Missing (n)	0	0	
BSI-18 (Anxiety)			
Baseline			0.72
Median (IQR)	0 (0–1)	0 (0–0)	
Mean (SD)	0.41 (0.76)	0.58 (1.33)	
Missing (n)	1	2	
Post-injury[1]			0.46
Median (IQR)	0 (0–2)	0 (0–2)	
Mean (SD)	1.17 (2.37)	1.31 (2.15)	
Missing (n)	10	3	
SCAT Total Symptom Severity			
Baseline			0.10
Median (IQR)	1 (0–3.25)	2 (0–5)	
Mean (SD)	2.57 (4.35)	4.98 (8.79)	
Missing (n)	1	1	
Post-injury[1]			0.01*
Median (IQR)	21 (11–40)	38 (23–54)	
Mean (SD)	27.33 (21.28)	38.53 (22.43)	
Missing (n)	0	0	
BSI-18 Total GSI Score			
Baseline			0.20
Median (IQR)	0 (0–2)	1 (0–2.5)	
Mean (SD)	1.41 (2.62)	1.54 (2.02)	
Missing (n)	1	2	
Post-injury[1]			0.20
Median (IQR)	3 (0.5–5)	5 (1.25–9)	
Mean (SD)	5.26 (7.20)	6.00 (6.00)	
Missing (n)	10	3	

Abbreviations: SRC, sport-related concussion; IQR, interquartile range; SD, standard deviation; SCAT, Sport Concussion Assessment Tool; BSI, Brief Symptom Inventory; GSI, Global Severity Index; MRI, magnetic resonance imaging.
† Fisher's exact test; Wilcoxon rank sum test.
* Statistically significant at $\alpha = 0.05$.
1 Post-injury scores collected <24 hours following SRC.

Table 3. MRI status on days to rtp-I and URTP.

	Hazard ratio	95% confidence interval	p-value
Days to Return to Play Initiation			
Symptom Severity	0.99	0.98–1.00	0.016*
MRI Status (No MRI = referent)[1]	0.75	0.28–1.21	0.23
Days to Return to Play Clearance			
Symptom Severity	0.98	0.97–0.99	<0.001*
MRI Status (No MRI = referent)[1]	0.77	0.29–1.26	0.30

* Statistically significant at $\alpha = 0.05$.
1 Multivariable Cox proportional hazards model adjusting for SCAT total symptom severity score (TSS).

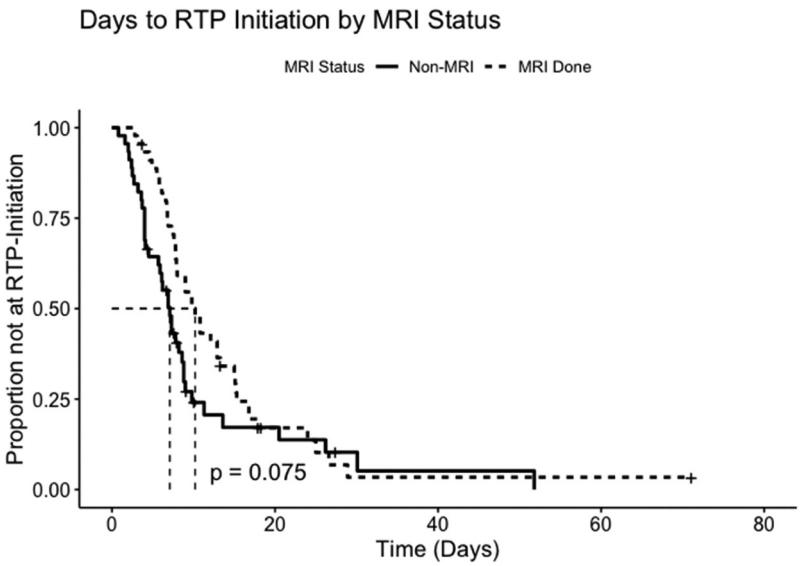

Figure 2. Kaplan-Meier plot for days to return to play (RTP) initiation by MRI status in 90 NCAA student-athletes.

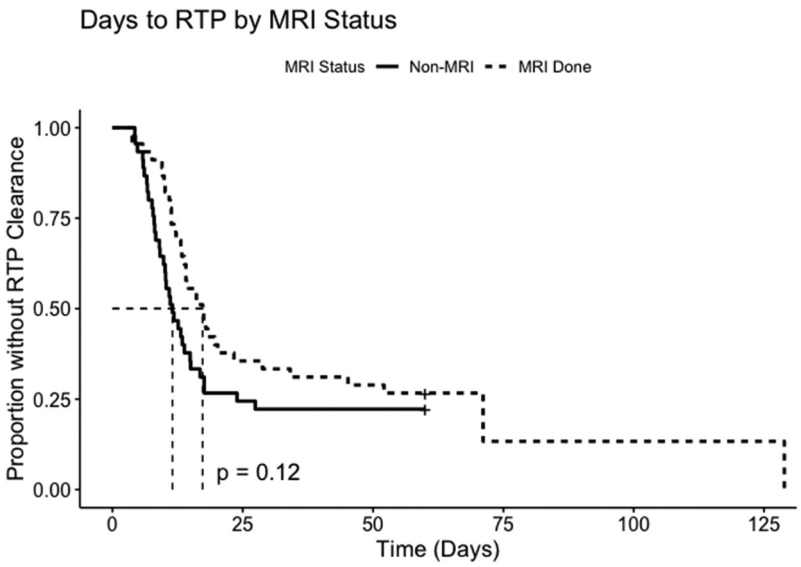

Figure 3. Kaplan-Meier plot for days to return to play (RTP) clearance by MRI status in 90 NCAA student-athletes.

symptom severity, we found that student-athletes undergoing MRI initiated the RTP protocol and had a total time to URTP comparable to the non-MRI group. Therefore, clinicians and researchers aiming to further explore an athlete's condition following injury need not worry that administering an MRI will prolong recovery and may include an MRI assessment without fear of impeding the recovery timeline.

During the post-injury phase (>24 h post-SRC), almost 40% of the student-athletes with MRI reported having some level of anxiety (38% on BSI-18; 36% on SCAT Nervous/Anxious), compared to less than 30% of the student-athletes who did not receive an MRI (29% on BSI-18, 27% on SCAT Nervous/Anxious). It is conceivable that increases in these symptoms have a biological relationship with the SRC but may also be related to psychological factors, such as uncertainty about future playing status (Delaney et al., 2015). Further, we observed increased total symptom burden (via SCAT TSS) in student-athletes post-SRC in the MRI versus non-MRI group (Wilcoxon rank-sum: *p-value* = 0.01), a difference which was not observed at baseline (*p-value* = 0.10). These symptom burden differences may have contributed to a difference in recovery between MRI groups, but such differences were not observed when controlling for TSS in the multivariable Cox model.

The present study did not evaluate the temporality of the MRI procedure (i.e., was symptom burden and/or anxiety assessed before or after MRI), making it unclear if student-athlete anxiety resulted from the SRC, the MRI procedure, or a combination of the two. The SCAT symptom inventory implemented in this investigation is the most widely used symptom questionnaire within sports medicine SRC research (Lempke et al., 2020, 2022), but it, and the BSI-18, is designed to evaluate state symptoms, which do not allow for an understanding of trait symptoms. State anxiety has a different biological underpinning from trait anxiety (Saviola et al., 2020) and future research may benefit from incorporating questionnaires specific to anxiety such as the State-Trait Anxiety Inventory at the time of MRI, to better understand this relationship (Beckler, 2010).

Outside of acute anxiety brought on by the MRI procedure, other investigators have reported MRI-induced vertigo, drowsiness, headache, and sleep disorders (Bravo et al., 2021; Roberts et al., 2011). This study was not designed to understand the relationship between these MRI-induced symptoms and SRC recovery but given how they are all similarly reported as SRC-related symptoms, future investigation is needed. Previous studies have elucidated cortical and subcortical changes associated with SRC in the CARE Consortium (Bobholz et al., 2021; Brett et al., 2020).

The present study did not account for the temporality of the MRI versus symptom evaluation; thus, it is possible that some student-athletes received the MRI before symptom evaluation or vice versa, possibly affecting anxiety or symptom self-reporting. Further, regardless of the order (e.g., MRI then symptom evaluation), it is possible that having both assessments conducted consecutively (i.e., symptom evaluation administered right before or after MRI) influenced symptom presence. Only 16 of 90 (18%) student-athletes in the present study were females (Table 1). Achieving a greater balance of a male-to-female student-athlete ratio would improve the generalizability of results. Further, the present study was limited to contact sport student-athletes, and future work examining MRI and SRC recovery will need to include non-contact sport student-athletes with SRC, as varying frequency and/or magnitude of head impacts may influence SRC recovery. Finally, the small sample size resultant of the propensity-score matching technique ($n = 90$ student-athletes from 2,072 eligible for analyses) limits generalizability but was necessary to reduce confounders to this research question. As the study included only NCAA student-athletes, it is unknown if the same findings translate to the youth population, as anxiety levels in elite young athletes have been examined in the years prior (Weber et al., 2018). Building on previous findings of limited cost-effectiveness (Morgan

et al., 2015), this study does not presently advise mandating MRI usage in SRC assessments but does support the idea that MRI can identify potentially significant cortical changes following SRC (Bobholz et al., 2021; Klein et al., 2019).

Conclusions

Our study found MRI use for research purposes did not significantly influence SRC recovery in NCAA student-athletes. We conclude that any uncomfortable feelings that may be experienced by a concussed athlete during such a procedure are short-lived and do not adversely affect recovery. Clinicians and researchers working with student-athletes therefore can be assured that the decision to undergo MRI will not produce lasting, deleterious effects.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This publication was made possible, in part, by support from the Grand Alliance CARE Consortium, funded by the National Collegiate Athletic Association and the Department of Defense. The United States Army Medical Research Acquisition Activity, 820 Chandler Street, Fort Detrick, MD 21702-5014, USA, is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under Award no. W81XWH-14-2-0151. Opinions, interpretations, conclusions, and recommendations are those of the authors and are not necessarily endorsed by the Department of Defense (Defense Health Program funds). The funders were not at all involved in the collection, analysis, interpretation, or reporting of the data presented in this manuscript.

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